

Additional examples of exam questions (including answers)

Question 1

The question

1) The simplest crop growth model originates from measurements showing that for many crops the potential growth rate amounts to $200 \text{ kg ha}^{-1} \text{ d}^{-1}$.

As you know, the Lintul model is based on the assumption that light intensity exponentially decreases with the accumulated LAI in a crop canopy, and that light can directly be related to the increase of biomass through the Light Use Efficiency (LUE).

Show that the Lintul model approximates the value of $200 \text{ kg ha}^{-1} \text{ d}^{-1}$ for a closed canopy, assuming that the light extinction coefficient, k , amounts to 0.6, the $\text{LUE}=3$, and the daily total radiation, DTR, equals $15 \text{ MJ m}^{-2} \text{ d}^{-1}$.

The answer A LAI of $3 \text{ m}^2 \text{ leaf m}^{-2}$ soil surface and a $k = 0.6$ mean that the term $(1.0 - e^{-(k \cdot \text{LAI})})$ in the equation $\text{PARINT} = 0.5 \cdot \text{DTR} \cdot (1.0 - e^{-(k \cdot \text{LAI})})$ amounts to 0.83(47), so PARINT is $0.5 \cdot \text{DTR} \cdot (1.0 - e^{-(k \cdot \text{LAI})}) = 0.5 \cdot 15 \cdot 0.83(47) = 6.26(03) \text{ MJ m}^{-2} \text{ d}^{-1}$, and the growth rate is given by $\text{GTOTAL} = \text{PARINT} \cdot \text{LUE}$, so $6.26(03) \cdot 3 = 18.78 \text{ g m}^{-2} \text{ d}^{-1}$ or $188 \text{ kg ha}^{-1} \text{ d}^{-1}$. This value favourably compares with the rule of thumb of $200 \text{ kg ha}^{-1} \text{ d}^{-1}$. (At a LAI of 4 the growth rate amounts to $205 \text{ kg ha}^{-1} \text{ d}^{-1}$.)

Question 2.

A research paper reports about data of two cultivars of spring barley. One result concerns development rates plotted versus temperature, Figure 1.

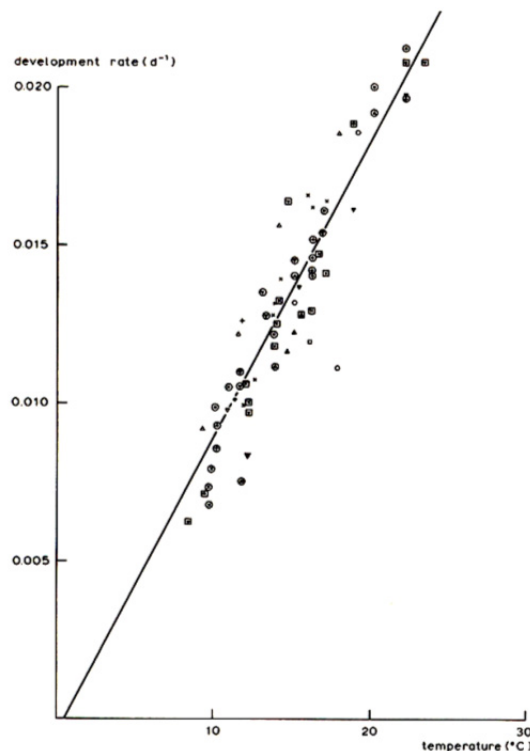


Figure 1. Development rate of barley as a function of temperature from emergence till flowering.

- 2a) Explain how the development rate, the values on the y-axis in this figure, is experimentally derived; also give the equation.

Development can be assessed by measuring the time elapsed between two distinct *crop development stages* at a series of constant temperatures in the greenhouse, for example.

The rate of development, $dDist/dt$, can then be derived using the equation:

$$\left\{ \frac{dDist}{dt} = \frac{\text{distance between 2 stages}}{\text{time period between these 2 stages}} \right\}_{T_a}$$

where “distance between 2 stages” may, for example, be the distance between emergence and flowering. Such a distance could be termed 1, or 100% or so, and does not have a unit. The “time period between these 2 stages”, the denominator, in days, must be assessed for a series of constant air temperatures, T_a . Plotting $dDist/dt$ versus T_a gives Figure 1.

- 2b) Calculate (in an accurate way, so using a ruler for example) from Figure 1 how long it takes in terms of physiological time, so expressed in °Cd, to reach flowering.

The slope of the line in Figure 1 has the unit $1/(^{\circ}\text{Cd})$, where °Cd represents the temperature sum or the heat sum needed to develop from emergence to flowering. It can be read from Figure 1 that $1/(^{\circ}\text{Cd}) = (0.020 - 0.005) / (22 - 6) = 9.375 \cdot 10^{-4}$, from which the TSUM can be calculated as $1 / (9.375 \cdot 10^{-4}) = 1067 ^{\circ}\text{Cd}$.

(Numbers between 1000 and 1200 will be accepted.)